Highway Traffic Volume Forecasting Based on Seasonal ARIMA Model*

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Abstract: In order to improve the accuracy of seasonal highway traffic volume forecasting, a general expression of seasonal ARIMA model with periodicity was presented based on the normal ARIMA model, and then the procedures of modeling and forecasting via seasonal ARIMA model were provided. In the feasibility-study experiment, the seasonal length was calculated by Fourier period analysis method, the test of the stationarity of the time series, and identifying, establishing, choosing and forecasting of the model were done by Eviews software. Compared with three normal seasonal forecasting models (grouped regression model, variable seasonal index forecasting model and seasonal regression model), the seasonal ARIMA model can obtain the highest accuracy in forecasting. The research result is significant to forecast highway traffic volume more accurately.

Key words: traffic engineering; highway transportation; traffic volume; seasonal ARIMA model; forecasting

0 Introduction

The scientific and accurate forecasting of highway transportation is not only the effective guarantee for the sustainable and healthy development of highway but also the important basis for the related decision department to making development strategy and planning. Traffic volume forecasting includes yearly forecasting and monthly forecasting. Because of the influence of climatic condition, holidays and people's customs and habits, and also because of the periodicity characteristic of the development of national economy, the monthly traffic volume has certain periodical feature. The forecasting effect of commonly used traffic volume forecasting method such as gray forecasting method[1], neural network method[2] and combination forecast method[3] is quite good in yearly forecasting, however the effect is not good when used in monthly forecasting.

A general expression of periodical ARIMA model was presented and the procedure of modeling and forecasting via this ARIMA model were provided. The result of feasibility study shows that the periodical ARIMA model obtained the highest accuracy in forecasting compared with other normal seasonal forecasting models.

1 ARIMA Model

ARIMA Model was put forward by American scholar Box and English scholoar Jenkins[4] in the sixties of 20th century. The model is also called Auto-regression Moving-average model, its basic idea is making the time series be stationary stochastic series by differential operational, then use the stationary stochastic series to build a model to describe the stochastic process, so we can use the model and observed value time series to calculate the best forecasting value corresponding to the future time series.

It is assumed that $\{x_t, t = 0, 1, \cdots\}$ is a stochastic time series, B is called backward shift operator, $Bx_t = x_{t-1}$. Denote $\nabla = 1 - B$, ∇ is difference operator. Then $\nabla x_t = (1 - B)x_t$, in general we can obtain:

$$\nabla^d x_t = (1 - B)^d x_t . \qquad (1)$$

If there is a nonnegative integer d to make formula (2) be tenabled, then $\{x_t\}$ is called autoregressive integration sliding average series, which is denoted as ARIMA(p, d, q), where d is difference order, p is autoregressive coefficient, q is moving average coefficient.

$$\phi(B)\nabla_d x_t = \theta(B)a_t , \qquad (2)$$

$$\phi(B) = 1 - \phi_1 B - \phi_2 B^2 - \cdots - \phi_p B^p , \qquad (3)$$

$$\theta(B) = 1 - \theta_1 B - \theta_2 B^2 - \cdots - \theta_q B^q , \qquad (4)$$

where $|B| \leqslant 1$, $\phi(B)$ and $\theta(B)$ are relatively prime, $\phi_p\theta_q \neq 0$, a_t is white noise series.

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2 Seasonal ARIMA model

In the practical problem, we often meet some stochastic series changed periodically with the time, the seasonal time series is obviously not stationary. However the data corresponding to the specific moment of each cycle is approximately ranged in same level, so if we use the observed data of one moment subtract the corresponding observed data of the same moment in next cycle, the periodical change may be eliminated, and the new series is approximately a stationary series[5].

Denote $\nabla_s = \mathrm{I} - B^s$, which is called seasonal difference operator, then

$$\nabla_s x_t = (1 - B^s) x_t = x_t - x_{t-s}, \qquad (5)$$

and we can obtain the D times seasonal difference

$$\nabla_s^D x_t = (\mathrm{I} - B^s)^D x_t = \nabla_s^{D-1} x_t - \nabla_s^{D-1} x_{t-s}, \qquad (6)$$

It is assumed that s is a positive integer, d and D are non-negative integers, if the stochastic series $\{x_t, t = 0, \mathrm{I}, \cdots\}$ meet the difference equation

$$\Phi(B^s) \nabla_s^D x_t = \Theta(B^s) \omega_t, \qquad (7)$$

then $\{x_t\}$ is seasonal ARIMA series, the order can be expressed by $(p, d, q) \times (P, D, Q)_s$,

$$\text{where, } \Phi(B^s) = \mathrm{I} - \Phi_1 B^s - \Phi_2 B^{2s} - \cdots - \Phi_P B^{Ps}, \qquad (8)$$

$$\Theta(B^s) = 1 - \Theta_1 B^s - \Theta_2 B^{2s} - \cdots - \Theta_Q B^{Qs}, \qquad (9)$$

among them, $\{\omega_t\}$ is ARIAM(p, d, q) series, that is

$$\phi(B) \nabla^d \omega_t = \theta(B) a_t, \qquad (\mathrm{I}0)$$

where, $\phi(B)$ and $\theta(B)$ meet the conditions of last segment.

By substitution of formula (10) for formula (7), the general seasonal ARIMA model can be obtained:

$$\phi_p(B) \Phi_P(B^s) \nabla^d \nabla_s^D x_t = \theta_q(B) \Theta_Q(B^s) a_t. \qquad (11)$$

3 Modeling process of seasonal ARIMA model

The steps of establishing the seasonal ARIMA model are[6,7]:

(I) Determine the period length s of the series by Fourier period analysis method.

(2) Identify the stationarity of the time series, that is to determine d and D, the common identifying tool is autocorrelogram. If the autocorrelation function tends to zero quickly, then we can say the time series is stationary time series. If the time series has some tendency, then difference to the time series is needed. If the time series has seasonal law, then seasonal difference to the time series is needed. If the time series has heteroscedasticity, then logarithmic transformation to the time series is needed.

(3) Identifying the model, that is to determine the order of the ARIMA model p, q, P and Q. In general, we can estimate the value of p, q, P and Q by observing autocorrelogram and partial autocorrelogram, then choose the best order value by AIC rule and SC rule.

(4) Obtain the estimated value of the all parameters of the model by maximum likelihood estimation.

(5) Forecasting.

In this paper, the modeling process is realized by the software Eviews3. I[8].

4 Empirical analysis

Tab. I shows the monthly vehicle flow statistical data of a highway from I99I to 2004. In order to make comparison between the actual value and the value from other forecasting models, the paper uses the data of former 13 years to forecast the monthly vehicle flow of 2004.

Tab.1 Highway traffic volume

Year	Month											
	1	2	3	4	5	6	7	8	9	10	11	12
1991	1 291	1 094	1 230	984	1 130	1 143	8 822	1 022	1 111	1 197	860	1 095
1992	1 234	1 390	1 251	1 070	1 270	1 228	978	1 090	1 092	1 321	1 090	1 246
1993	1 507	1 488	1 356	1 119	1 259	1 283	1 084	1 214	1 253	1 276	1 221	1 267
1994	1 609	1 230	1 202	1 141	1 335	1 258	1 039	1 174	1 164	1 245	1 209	1 235
1995	1 325	1 546	1 352	1 242	1 365	1 331	1 162	1 124	1 103	1 184	1 068	1 088
1996	1 383	1 402	1 162	1 161	1 313	1 310	1 140	1 138	1 102	1 373	1 184	1 354
1997	1 688	1 211	1 526	1 239	1 455	1 387	1 198	1 259	1 272	1 310	1 188	1 239
1998	1 456	1 776	1 326	1 280	1 346	1 273	1 085	1 216	1 207	1 296	1 231	1 280
1999	1 584	1 426	1 291	1 202	1 261	1 140	1 116	1 160	1 231	1 317	1 201	1 311
2000	1 305	1 767	1 359	1 234	1 312	1 337	1 286	1 502	1 523	1 563	1 444	1 623
2001	1 853	1 757	1 415	1 332	1 424	1 432	1 326	1 484	1 513	1 540	1 451	1 601
2002	2 030	1 687	1 548	1 507	1 553	1 424	1 338	1 419	1 474	1 546	1 466	1 582
2003	1 698	2 109	1 792	1 603	1 632	1 571	1 453	1 770	1 608	1 554	1 599	1 695
2004	1 939	2 018	1 686	1 589	1 687	1 610	1 623	1 974	1 877	1 876	1 863	1 934

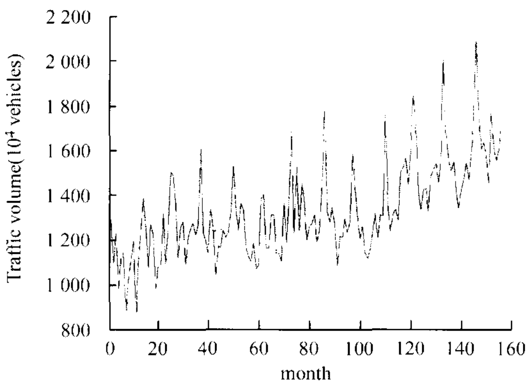

Fig.1 Time series of highway traffic volume

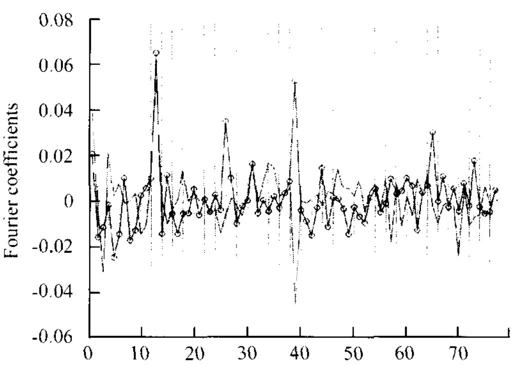

Fig.2 Change curve of Fourier coefficient

4.1 Seasonal identification

Drawing the time series figure (Fig. 1), we can see the time series has linear growth trend, and approximately shows periodicity. So, the paper uses Fourier period analysis method to determine the seasonal length. Given a time series x, and we can obtain the new series x_t if the tendency is removed. Here t is understood as continuous variable, and the value range is $[0, n]$, then x_t is a smooth continuous function $x(t)$ add stochastic item u_t, that is $x_t = x(t) + u_t$. It is assumed that $x(t)$ meet all of the Fourier expansion conditions, expansion the function $x(t)$ to Fourier's series in the range $[-n/2, n/2]$:

$$x(t) = \sum_{k=0}^{m} (a_k \cos \frac{2k\pi}{n} t + b_k \sin \frac{2k\pi}{n} t), \quad (12)$$

where, m is the maximum integer no more than $n/2$. The coefficients are obtained by least square method:

$$a_0 = \frac{1}{n} \sum_{t=1}^{n} x_t, \ a_k = \frac{2}{n} \sum_{t=1}^{n} x_t \cos \frac{2k\pi}{n} t,$$

$$b_k = \frac{2}{n} \sum_{t=1}^{n} x_t \sin \frac{2k\pi}{n} t, \ k = 1, 2, \cdots, m, \quad (13)$$

If the coefficients a_k and b_k are all very small except a_0, then x_t is a constant, that is to say there is no seasonality. If a pair of coefficients a_k and b_k are some larger, then there is seasonal composition and the period length is $L = \frac{n}{k}$.

In the example, the tendency function is obtained by least square method:

$$x_t = 1\,093.6 + 3.1t, \quad (14)$$

then the Fourier coefficients a_k b_k can be computed. Fig. 2 shows the variation curve. From the figure we can see, when $k = 13$, $k = 26$, and $k = 39$, a_k and b_k are some larger. Here $n = 156$, so we can say the tome series has seasonal composition and period length is $L = n/13 = 12$.

4.2 Stationarity test of the time series

In order to eliminate the tendency and reduce the volatility, do first order natural logarithm difference to the original series x_t, the Eviews command is:

$$\text{series } ilx_t = \ln(x_t) - \ln(x_t(-1)). \quad (15)$$

From the autocorrelogram and partial autocorrelogram of the new series (Fig. 3), we can see the tendency of the series is nearly eliminated, however the autocorrelation coefficient and partial autocorrelation coefficient are not equal to 0 obviously when $k = 12$, that is to say the seasonality is exiting. Do seasonal difference to the series ilx_t, the new series $silx_t$ is obtained:

$$\text{seies } sil_x = ilx_t - ilx_t(-12). \quad (16)$$

In order to test the forecasting effect of the model, set aside the observation value of 2003 as the reference for evaluation of forecast accuracy, then the sample period of modeling is from January 1991 to December 2002.

Drawing the auto autocorrelogram and partial autocorrelogram of series $silx_t$ (Fig. 4), we can see the autocorrelation coefficient and partial autocorrelation coefficient fall into random interval quickly, which means the tendency has been nearly eliminated. However, when $k = 12$, the autocorrelation coefficient and the partial autocorrelation coefficient are still relatively great, the seasonality is still quite obvious. By doing experiment, the seasonality cannot be eliminated by second order seasonal difference, so we do first order difference to the series.

Do ADF test to the series $silx_t$, the statistical value of t is equal to -7.71, which is smaller than the critical value -3.48 when significance level is one percent, that is to say the series is reposeful, so we can establish the ARIMA model.

4.3 Identification of model

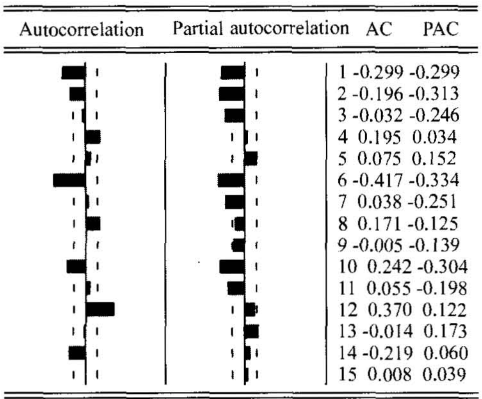

Fig.3 Autocorrelation-partial autocorrelation analysis of series ilx_t

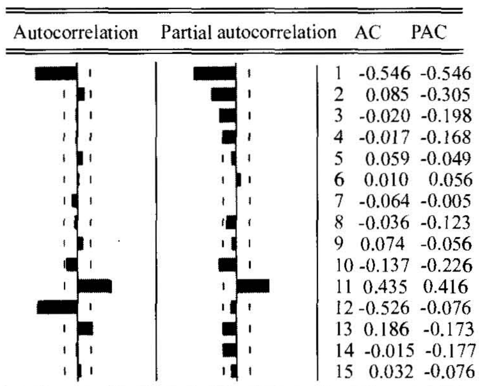

Fig.4 Autocorelation-partial autocorrelation analysis of series $silx_t$

In the example, $d = 1$, $D = 1$. From the partial autocorrelogram, p can be equal to 2 or 3. From the autocorrelogram q is equal to 1.

In the practical application, the AR model is more popular used compared to MA model and ARMA model, because the parameters are easier to explain. By comprehensive consideration, the combination of (p, q) can be (3,1), (4,0), (2, 1) and (3, 0). Because the autocorrelation coefficient and partial autocorrelation coefficient are not equal to 0 obviously when k is equal to 12, so P and Q are all equal to 1.

4.4 Establishment and selection of model

Establish ARIMA$(3,1,1)(1,1,1)^{1/2}$ model, and use it to trial forecast the data of 2003, then establish the other 3 models similarly, the 4 models' test result aggregation is shown in Tab.2.

Tab.2 Test result of each model

(p, q)	AIC	SC	MAPE
(3,1)	-2.286	-2.173	6.385
(2,1)	-2.316	-2.189	6.217
(4,0)	-2.283	-2.176	6.794
(3,0)	-2.299	-2.171	6.588

From Tab.2, we can see that the AIC value, SC value and trial forecasting MAPE are all smallest, so we choose ARIMA$(2,1,1)(1,1,1)^{1/2}$.

4.5 Forecasting

The estimated values of all the parameters of the ARIMA$(2, 1, 1)(1, 1, 1)^{1/2}$ model are obtained by maximum likelihood estimation method, the expansion equation of the model is:

$$(1+0.229B^{12})(1+0.470B+0.198B^2)(1-B)(1-B^{12})\ln(x_t) = (1+0.122B)(1+0.882B^{12})a, \quad (17)$$

The forecast result of monthly vehicle flow in 2004 is shown in Tab.3.

5 Comparative analysis

In order to be compared with the forecasting model built by the paper, the commonly used seasonal forecasting models such as grouped regression model, variable seasonal index model and seasonal regression model are built. The forecasting results can be seen in Tab.3.

Tab.3 Forecasting result of each model

Month	1	2	3	4	5	6	7	8	9	10	11	12	MAPE
Actual values	1 939	2 018	1 686	1 589	1 687	1 610	1 623	1 974	1 877	1 876	1 863	1 934	-
ARIMA model	1 982	1 934	1 710	1 577	1 731	1 669	1 488	1 595	1 635	1 752	1 610	1 735	0.071 4
grouped regression model	1 835	1 929	1 586	1 511	1 544	1 469	1 422	1 596	1 565	1 574	1 565	1 669	0.110 4
variable seasonal index model	1 817	1 936	1 566	1 511	1 530	1 450	1 426	1 596	1 559	1 559	1 573	1 667	0.134 1
seasonal regression model	1 727	1 705	1 595	1 592	1 677	1 655	1 544	1 541	1 626	1 604	1 494	1 491	0.110 6

From the Tab.3 we can see that the forecasting error of ARIMA model build by the paper is the smallest, that is to say the model can obtain the best seasonal forecasting performance.

6 Conclusion

In this paper, the way of using seasonal ARIMA model to forecast the highway traffic volume is presented, and an empirical analysis is also given. The result shows that the model is feasible and reliable. However the forecasting method is completely data driven, it has a certain limitation. So the problem about binding qualitative analysis is a topic which still needs further research.

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